Venkata Lakshmi Parimala Pasupuleti

Developer Intelligence · Platform Engineering · AI-Assisted SDLC · CI/CD · Observability · Go · Python · LLM Integration

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PROFESSIONAL SUMMARY

I build AI-powered developer intelligence systems - telemetry pipelines, automated code review, LLM-assisted incident triaging - with 9+ years of production engineering (7+ years at Entrata, a PropTech SaaS platform for 100K+ MAU). Currently at Intuit integrating Anthropic Claude into financial applications, with Go-based telemetry collection services and observability instrumentation for LLM workflow pipelines. Built AgentProbe, an open-source Developer Intelligence Toolkit: a CI regression harness with LLM-as-judge evaluation for automated software validation across 1,240+ test cases. Expert in Python (9+ yrs), AI-assisted SDLC workflows, CI/CD pipeline integration, and cloud infrastructure (AWS, GCP, Azure). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

CORE COMPETENCIES

Developer Intelligence | Engineering Telemetry | Platform Engineering | AI-Assisted SDLC | LLM Integration | Observability (Logs/Metrics/Traces) | CI/CD Pipelines | Go | Python (9+ yrs) | Automated Code Review | Incident Triaging | Developer Velocity | Feedback Loops | Workflow Orchestration | AWS / GCP / Azure | Data Pipelines | LangGraph / LangChain | OpenTelemetry

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer - AI Platform & Developer Tooling · Virginia, USA

- **Developed Go-based telemetry collection services** for LLM workflow observability, integrating with CI/CD pipelines to surface developer intelligence signals - engineering telemetry streams for automated feedback loops across the SDLC.
- **Instrumented distributed LLM systems with structured logging, metrics, and trace collection** (OpenTelemetry, Prometheus, Grafana) for production incident triaging; dashboards surface quality regressions before user impact.
- Integrated Anthropic Claude into Python/FastAPI services for financial data summaries and natural-language queries; designed structured-output schemas for consistent, validated downstream consumption.
- Built **LLM-as-judge evaluation and golden-dataset regression tests** gated on GitHub Actions CI/CD - automated software validation that catches hallucinations and quality drops at commit time.
- Added guardrails around LLM calls: input validation, sensitive-data handling, and fallbacks; built **data pipelines** from ingestion through feature engineering to prediction APIs on AWS.
- Containerized services with Docker; deployed through **GitHub Actions CI/CD to AWS** with production logging and monitoring for full observability.
- Grounded Claude responses in customer financial data using retrieval pipelines so answers are traceable to source records; built ML pipelines with scikit-learn and XGBoost.

Entrata - Apartment & Affordable Housing SaaS

Oct 2020 - Jul 2024

Senior Software Engineer · **Multi-tenant platform, 100K+ monthly active users**

- **Platform engineering at scale:** delivered 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints on a production SaaS platform with 100K+ MAU.
- **CI/CD ownership and developer tooling:** worked on a Docker microservices redesign (50% better scalability, 30% faster deployments); owned the AWS S3 document migration with Libsodium encryption for resident PHI - zero security incidents.
- Wrote **Python/Bash automation that cut manual operational work by 80%**; built internal debugging tools that reduced developer triage time by 40% - direct developer velocity impact.
- Optimized 200+ SQL queries using **execution plans and indexing**, improving performance by 30%; maintained the HUD income-limit import pipeline for 6 years (180K domain records, zero compliance violations).
- Built a **RabbitMQ-based async processing system** with retry logic and idempotency for rent updates across thousands of tenant records - production-grade data pipeline engineering.
- **Reviewed 300+ pull requests, mentored junior engineers** on platform architecture - active knowledge transfer and code quality ownership.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · **Pune, India**

- Built property management modules (Python, PHP, Java, PostgreSQL) for affordable housing and rental management.
- Delivered large-scale data import and reconciliation workflows (1.6M HUD/Tax Credit records); led AWS S3 cloud migration from legacy server storage.

PROJECTS

AgentProbe - Developer Intelligence Toolkit

Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions CI/CD · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama · github.com/venkatalakshmiparimala/AgentProbe

Built an open-source **automated code quality evaluation and CI regression harness** - a 15-endpoint FastAPI backend routing prompts across five LLM providers with LLM-as-judge scoring for factual accuracy, hallucination rate, retrieval quality, completeness, and compliance. **Automated software validation:** CI-integrated golden dataset regression tests catch quality drops at commit time. Scores LLM outputs across 1,240+ test cases (93.6% precision, 87.9% F1). Three LangGraph agents: Orchestrator, RAG Tracer, Auto Fix - automated developer intelligence feedback loops for the SDLC.

Leasel - AI-Assisted Workflow Orchestration Platform

LangGraph · 8-Agent

Python · LangGraph (8-agent StateGraph) · FastAPI · Hybrid Retrieval (TF-IDF + semantic) · OCR · SQLite · CI/CD · github.com/venkatalakshmiparimala/Lease_complaince_rag

Smart workflow orchestration for compliance operations: an 8-agent LangGraph StateGraph (triage, retrieval, reasoning, validation) that automates manual developer-facing compliance checks with conditional routing and human escalation when confidence is low. Hybrid-search RAG pipeline (TF-IDF + semantic reranking) with answers cited to exact source clauses. **CI-integrated LLM-as-judge harness**: 93.6% precision across 1,240 test cases; safety boundary enforced in code - eligibility decisions always route to staff.

AI on FHIR - Clinical NLP & LLM API Integration

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

Built an app that turns English/Spanish questions into FHIR API calls using LLMs, BERT, and embeddings, cutting manual clinical lookup time by ~60%. Role-based access control for HIPAA-style data isolation; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering - George Mason University**, Virginia** May 2026

GPA 3.87 · Coursework: ML pipelines, NLP, cloud systems, data engineering, custom classification models (PyTorch + HuggingFace)

B.Tech Computer Science & Engineering - JNTU Anantapur 2017

TECHNICAL SKILLS

Languages & Platform: Go, Python (9+ yrs), Java, Spring Boot, TypeScript, JavaScript, PHP, Kotlin, Scala, Bash, C#

AI & LLM Integration: Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, LangGraph, LangChain, RAG Pipelines, LLM-as-Judge, AI-Assisted Workflows, Prompt Engineering, MCP

Developer Intelligence & Observability: Engineering Telemetry, OpenTelemetry, Prometheus, Grafana, Logs/Metrics/Traces, Incident Triaging, Developer Metrics Frameworks, SDLC Feedback Loops

CI/CD & Platform: GitHub Actions, Docker, Kubernetes, Terraform, Microservices, Developer Velocity, Software Catalog, Automated Code Review, Software Validation

Cloud & Data Pipelines: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway), Azure (OpenAI, Databricks, AKS), GCP (Vertex AI), ETL Pipelines, Data Modeling, Kafka, RabbitMQ

Databases & ML: PostgreSQL, MySQL, Oracle, Redis, MongoDB, Snowflake, ChromaDB, Pinecone, Elasticsearch, scikit-learn, XGBoost, PyTorch, HuggingFace, spaCy